Supplementary Material for the Paper "ECA-BLS: An Efficient Complex-Augmented Broad Learning System"

S.I. Related Works

This section provides a concise yet comprehensive review of the Broad Learning System (BLS), together with its mathematical formulation.

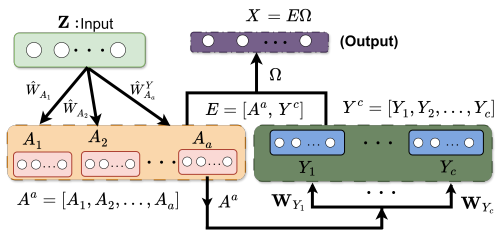


Fig. 1: Architecture of the Broad Learning System (BLS).

A. Broad Learning System (BLS)

The Broad Learning System (BLS), proposed by Chen *et al.*, is a flat network architecture designed to achieve fast learning with strong generalization capability. Unlike deep neural networks, BLS expands the network width rather than depth and consists of four main components: the input layer, feature learning layer, enhancement layer, and output layer. The connections between successive layers are randomly generated, while the output weights are computed analytically using a closed-form solution. The overall architecture of BLS is illustrated in Fig. 1. The mathematical formulation of each component is described below.

1) Feature Learning Segment: Assume that the feature learning layer is composed of a feature groups, each containing b nodes. The output of the ith feature group is defined as

$$\mathbf{A}_i = \sigma_i\big(\mathbf{Z}\mathbf{W}_{A_i} + \boldsymbol{\xi}_{A_i}\big) \in \mathbb{R}^{D\times b}, \quad i = 1, 2, \ldots, a, \tag{1}$$

where $\sigma_i(\cdot)$ denotes the nonlinear activation function, $\mathbf{W}_{A_i} \in \mathbb{R}^{B\times b}$ is a randomly generated weight matrix, and $\boldsymbol{\xi}_{A_i} \in \mathbb{R}^{D\times b}$ is the corresponding bias matrix. The outputs of all feature groups are concatenated to form the augmented feature matrix

$$\mathbf{A}^a = [\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_a] \in \mathbb{R}^{D\times ab}. \tag{2}$$

2) Enhancement Segment: To further enrich the feature representation, the augmented feature matrix $\mathbf{A}^a$ is projected into an enhancement layer through random mappings followed by nonlinear activation. Let c denote the number of enhancement groups, with each group consisting of d nodes. The output of the kth enhancement group is given by

$$\mathbf{Y}_k = \zeta_k(\mathbf{A}^a\mathbf{W}_{Y_k} + \boldsymbol{\Gamma}_{Y_k}) \in \mathbb{R}^{D\times d}, \quad k = 1, 2, \ldots, c, \tag{3}$$

where $\zeta_k(\cdot)$ is a nonlinear activation function, $\mathbf{W}_{Y_k} \in \mathbb{R}^{ab\times d}$ is a randomly generated weight matrix, and $\boldsymbol{\Gamma}_{Y_k} \in \mathbb{R}^{D\times d}$ is the bias matrix. The outputs of all enhancement groups are concatenated to form

$$\mathbf{Y}^c = [\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_c] \in \mathbb{R}^{D\times cd}. \tag{4}$$

3) Output Segment: The output layer receives the concatenation of the feature and enhancement matrices. The network output is computed as

$$\mathbf{O} = [\mathbf{A}^a, \mathbf{Y}^c]\,\boldsymbol{\Omega} = \mathbf{E}\boldsymbol{\Omega}, \tag{5}$$

where

$$\mathbf{E} = [\mathbf{A}^a, \mathbf{Y}^c] \in \mathbb{R}^{D\times(ab+cd)} \tag{6}$$

is the final hidden-layer output matrix, and $\boldsymbol{\Omega} \in \mathbb{R}^{(ab+cd)\times C}$ denotes the output weight matrix. The optimal output weights are obtained analytically using the Moore–Penrose pseudoinverse as

$$\boldsymbol{\Omega} = \mathbf{E}^{\dagger}\mathbf{X}, \tag{7}$$

where $\mathbf{E}^{\dagger}$ represents the pseudoinverse of $\mathbf{E}$.

S.II Theoretical Proof on the Equivalence of the Proposed CA-BLS and ECA-BLS

Theorem .1. *Suppose the training samples are distinct and the same hidden representation is used. Let β_a denote the output weight matrix obtained from CA-BLS with regularization parameter λ_a. Let β_r denote the output weight matrix obtained from ECA-BLS with regularization parameter λ_r. If $\lambda_a = 2\lambda_r$, then the outputs of the two models are identical, i.e.,*

$$\mathbf{H}_a\beta_a = \mathbf{H}_r\beta_r. \tag{8}$$

Proof. Let $\mathbf{H} \in \mathbb{C}^{N\times M}$ be the complex-valued hidden-layer output matrix.

a) Augmented representations.: The CA-BLS augmented matrix is defined as

$$\mathbf{H}_a = [\,\mathbf{H},\, \mathbf{H}^*\,] \in \mathbb{C}^{D\times 2(ab+cd)}, \tag{9}$$

while the ECA-BLS real augmented matrix is given by

$$\mathbf{H}_r = [\,\Re(\mathbf{H}),\, \Im(\mathbf{H})\,] \in \mathbb{R}^{D\times 2(ab+cd)}. \tag{10}$$

b) Linear transformation between augmented matrices.: Define the transformation matrix

$$\mathbf{V} = \begin{bmatrix} \mathbf{I} & \mathbf{I} \\ i\mathbf{I} & -i\mathbf{I} \end{bmatrix} \in \mathbb{C}^{2(ab+cd)\times 2(ab+cd)}, \tag{11}$$

where $\mathbf{I}$ denotes the $M \times M$ identity matrix. Using $\mathbf{H} = \Re(\mathbf{H}) + i\Im(\mathbf{H})$ and $\mathbf{H}^* = \Re(\mathbf{H}) - i\Im(\mathbf{H})$, it follows that

$$\mathbf{H}_a = \mathbf{H}_r\mathbf{V}. \tag{12}$$

It is straightforward to verify that

$$\mathbf{V}\mathbf{V}^H = 2\mathbf{I} \implies \mathbf{V}^{-1} = \frac{1}{2}\mathbf{V}^H. \tag{13}$$

c) Output weight solutions.: The CA-BLS output weights are computed as

$$\beta_a = (\mathbf{H}_a^H\mathbf{H}_a + \lambda_a\mathbf{I})^{-1}\mathbf{H}_a^H\mathbf{x}. \tag{14}$$

Substituting $\mathbf{H}_a = \mathbf{H}_r\mathbf{V}$ and using the property $\mathbf{H}_r^H = \mathbf{H}_r^T$ (since $\mathbf{H}_r$ is real), we obtain:

$$\beta_a = (\mathbf{V}^H\mathbf{H}_r^T\mathbf{H}_r\mathbf{V} + \lambda_a\mathbf{I})^{-1}\mathbf{V}^H\mathbf{H}_r^T\mathbf{x}. \tag{15}$$

To simplify the inverse term, we use the identity $\mathbf{I} = \frac{1}{2}\mathbf{V}^H\mathbf{V}$ derived from (13). We substitute $\lambda_a\mathbf{I} = \frac{\lambda_a}{2}\mathbf{V}^H\mathbf{V}$:

$$\beta_a = \left(\mathbf{V}^H\mathbf{H}_r^T\mathbf{H}_r\mathbf{V} + \mathbf{V}^H\left(\frac{\lambda_a}{2}\mathbf{I}\right)\mathbf{V}\right)^{-1}\mathbf{V}^H\mathbf{H}_r^T\mathbf{x}. \tag{16}$$

Factoring out $\mathbf{V}^H$ and $\mathbf{V}$:

$$\beta_a = \left[\mathbf{V}^H\left(\mathbf{H}_r^T\mathbf{H}_r + \frac{\lambda_a}{2}\mathbf{I}\right)\mathbf{V}\right]^{-1}\mathbf{V}^H\mathbf{H}_r^T\mathbf{x}. \tag{17}$$

Using the inverse product property $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$:

$$\beta_a = \mathbf{V}^{-1}\left(\mathbf{H}_r^T\mathbf{H}_r + \frac{\lambda_a}{2}\mathbf{I}\right)^{-1}(\mathbf{V}^H)^{-1}\mathbf{V}^H\mathbf{H}_r^T\mathbf{x}. \tag{18}$$

Simplifying using $(\mathbf{V}^H)^{-1}\mathbf{V}^H = \mathbf{I}$ and $\mathbf{V}^{-1} = \frac{1}{2}\mathbf{V}^H$:

$$\beta_a = \frac{1}{2}\mathbf{V}^H\left(\mathbf{H}_r^T\mathbf{H}_r + \frac{\lambda_a}{2}\mathbf{I}\right)^{-1}\mathbf{H}_r^T\mathbf{x}. \tag{19}$$

The ECA-BLS solution with regularization parameter λ_r is defined as:

$$\beta_r = \left(\mathbf{H}_r^T\mathbf{H}_r + \lambda_r\mathbf{I}\right)^{-1}\mathbf{H}_r^T\mathbf{x}. \tag{20}$$

Comparing this with (19), if we set $\lambda_r = \frac{\lambda_a}{2}$ (or equivalently $\lambda_a = 2\lambda_r$), we establish the relationship:

$$\beta_a = \frac{1}{2}\mathbf{V}^H\beta_r. \tag{21}$$

d) Equivalence of outputs.: Finally, the CA-BLS output is calculated as:

$$\mathbf{H}_a\beta_a = (\mathbf{H}_r\mathbf{V})\left(\frac{1}{2}\mathbf{V}^H\beta_r\right). \tag{22}$$

Using $\mathbf{V}\mathbf{V}^H = 2\mathbf{I}$, we obtain:

$$\mathbf{H}_a\beta_a = \mathbf{H}_r\left(\frac{1}{2}(2\mathbf{I})\right)\beta_r = \mathbf{H}_r\beta_r. \tag{23}$$

This completes the proof. □

S.III Algorithm of proposed ECA-BLS

The algorithm of our proposed model is:

Algorithm 1 Algorithm of the proposed ECA-BLS

Input: Input matrix $\mathbf{Z} \in \mathbb{R}^{D\times B}$, target matrix $\mathbf{x} \in \mathbb{R}^{D\times C}$, where the dataset consists of D samples with B features and C classes.
Parameters: λ_r, a , b, c , d.
Transform: Map real-valued input to complex domain using phase encoding ${}^{cx}\mathbf{Z} = \exp(j\pi\mathbf{Z})$.
Find: Feature node outputs ${}^{cx}\mathbf{A}_i \in \mathbb{C}^{D\times b}$, $i = 1, 2, \ldots, a$, using Eq. 2 of the main manuscript.
Find: Concatenated feature matrix ${}^{cx}\mathbf{A}^a = [{}^{cx}\mathbf{A}_1, {}^{cx}\mathbf{A}_2, \ldots, {}^{cx}\mathbf{A}_a] \in \mathbb{C}^{D\times ab}$.
Find: Enhancement node outputs ${}^{cx}\mathbf{Y}_k \in \mathbb{C}^{D\times d}$ using Eq. 5 of the main manuscript, $k = 1, 2, \ldots, c$, using random orthogonal complex weights.
Find: Concatenated enhancement matrix ${}^{cx}\mathbf{Y}^c = [{}^{cx}\mathbf{Y}_1, {}^{cx}\mathbf{Y}_2, \ldots, {}^{cx}\mathbf{Y}_c] \in \mathbb{C}^{D\times cd}$.
Find: Hidden-layer output matrix $\mathbf{H} = [{}^{cx}\mathbf{A}^A, {}^{cx}\mathbf{Y}^c] \in \mathbb{C}^{D\times(ab+cd)}$.
Construct: Real-augmented hidden matrix $\mathbf{H}_r = [\Re(\mathbf{H}), \Im(\mathbf{H})] \in \mathbb{R}^{D\times 2(ab+cd)}$.
Output: Compute output weight matrix

$$\boldsymbol{\beta}_r = (\mathbf{H}_r^T\mathbf{H}_r + \lambda_r\mathbf{I})^{-1}\mathbf{H}_r^T\mathbf{X}.$$

S. IV Complexity analysis of proposed models

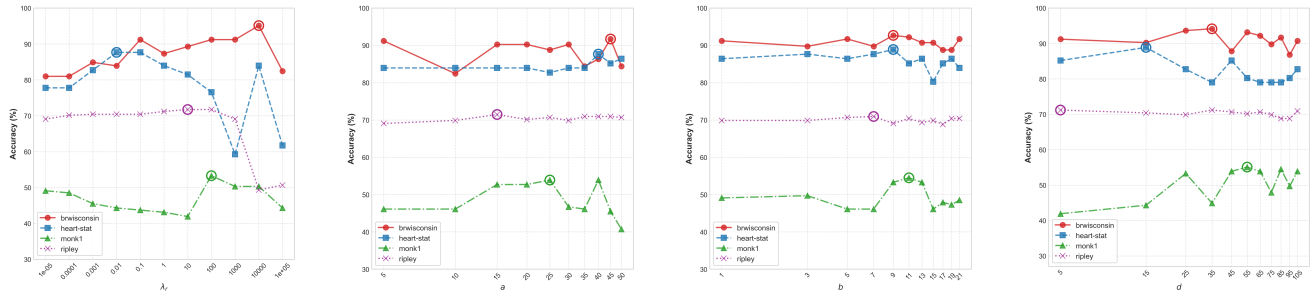
For ECA-BLS, the augmented hidden matrix $\mathbf{H}_r \in \mathbb{R}^{D\times 2(ab+cd)}$ is real-valued. The computation of $\mathbf{H}_r^T\mathbf{H}_r$ requires $D \times 2(ab + cd) \times 2(ab + cd)$ real multiplications, leading to a quadratic term of order $\mathcal{O}\big(8D(ab+cd)^2\big)$. Subsequently, inverting Eq. 12 of the main manuscript resulting $2(ab + cd) \times 2(ab + cd)$ matrix incurs a cubic cost of $\mathcal{O}\big(8(ab+cd)^3\big)$. Finally, the multiplication $\mathbf{H}_r^T\mathbf{x}$ contributes $\mathcal{O}(2D(ab+cd)C)$ additional real multiplications. Together, these operations yield an overall training complexity of $\mathcal{O}\big(8(ab+cd)^3 + 8D(ab+cd)^2 + 2D(ab+cd)C\big)$, with the same asymptotic order for addition operations.

In contrast, CA-BLS operates on an complex augmented hidden matrix $\mathbf{H}_a \in \mathbb{C}^{D\times 2(ab+cd)}$ and employs Hermitian products and complex-valued matrix inversion. The computation of $\mathbf{H}_a^H\mathbf{H}_a$ involves complex multiplications, each equivalent to four real multiplications and two real additions. Consequently, the quadratic term increases to $\mathcal{O}\big(32D(ab+cd)^2\big)$ real multiplications and $\mathcal{O}\big(24D(ab+cd)^2\big)$ real additions. Similarly, inverting the $2(ab + cd) \times 2(ab + cd)$ complex matrix in Eq. 10 of the main manuscript incurs a cubic cost of $\mathcal{O}\big(32(ab+cd)^3\big)$ real multiplications and $\mathcal{O}\big(24(ab+cd)^3\big)$ real additions. The computation of $\mathbf{H}_a^H\mathbf{x}$ further contributes $\mathcal{O}(8D(ab+cd)C)$ real multiplications and $\mathcal{O}(6D(ab+cd)C)$ real additions.

TABLE I: Pairwise win-tie-loss test of proposed models and baseline models on MCD category UCI datasets.

↓ Model →	**BLS** [1]	**H-ELM** [2]	**GEIB** [3]	**F-BLS** [4]	**IF-BLS** [4]	**KRP-BLS** [5]
H-ELM [2]	[11, 1, 14]					
GEIB [3]	[13, 0, 13]	[15, 1, 10]				
F-BLS [4]	[11, 0, 15]	[13, 1, 12]	[11, 0, 15]			
IF-BLS [4]	[11, 3, 12]	[16, 0, 10]	[14, 1, 11]	[15, 1, 10]		
KRP-BLS [5]	[10, 2, 14]	**[18, 0, 8]**	[12, 1, 13]	**[18, 0, 8]**	[10, 2, 14]	
ECA-BLS†	[17, 1, 8]	**[20, 1, 5]**	[17, 2, 7]	**[20, 0, 6]**	[14, 2, 10]	[14, 2, 10]

† represents the proposed models.



(a) (b) (c) (d)

Fig. 2: Performance variation of the ECA-BLS model with respect to parameters λ_r, a, b and d, respectively.

Compared to CA-BLS, ECA-BLS reduces the dominant cubic multiplication complexity from $32n^3$ to $8n^3$, achieving an exact reduction of 75%. Likewise, the dominant cubic addition complexity decreases from $24n^3$ to $8n^3$, corresponding to an approximate reduction of 66.7%. Since the cubic term dominates the overall training cost, ECA-BLS achieves an effective fourfold reduction in computational complexity while preserving an identical decision function, as guaranteed by the theoretical equivalence analysis.

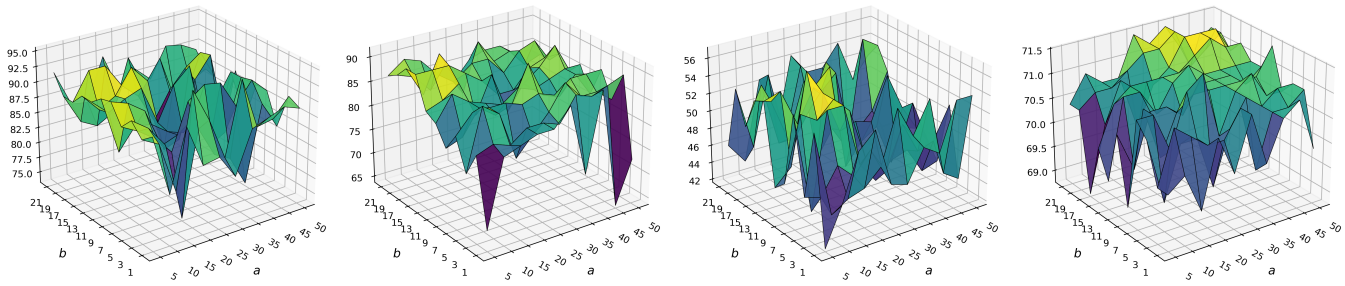
S.V. Experimental Setup, Compared Models and Datasets

Experimental Setup: The experimental configuration includes a PC equipped with an Intel(R) Xeon(R) Gold 6226R CPU running at a speed of 2.90 GHz and featuring 128 GB of RAM. The system runs on the Windows 11 platform and executes tasks using Python 3.11. The dataset is randomly divided into training and testing subsets in a ratio of 70:30, respectively. To tune the model hyperparameters, five-fold cross-validation is employed, followed by a comprehensive grid search over predefined parameter ranges. The regularization parameter λ_r is selected from $\{10^{-5}, 10^{-4}, \ldots, 10^{5}\}$, Additionally, a denotes the number of feature groups, b the number of nodes within each feature group, d the number of enhancement nodes in each enhancement group, and c the total number of enhancement groups. The parameter a is selected from $\{5, 10, 15, \ldots, 50\}$, b from $\{1, 3, 5, \ldots, 21\}$, d from $\{5, 15, 25, \ldots, 105\}$, and c is fixed to 1. The ECA-BLS network utilizes the holomorphic inverse hyperbolic sine activation function, known as $\operatorname{arcsinh}(\cdot)$.

Compared Models: To rigorously assess the effectiveness of the proposed models, we perform a comprehensive experimental comparison against a diverse set of representative random deep and broad learning (RdNN)-based architectures. The selected benchmark methods are systematically categorized to reflect different design philosophies and robustness mechanisms, including the classical Broad Learning System (BLS) [1], Hierarchical Extreme Learning Machine (H-ELM) [2], fuzzy Broad Learning System (F-BLS) [4], intuitionistic fuzzy Broad Learning System (IF-BLS) [4], graph-embedding intuitionistic BLS (GEIB) [3], and kernel risk-sensitive mean p-power BLS (KRP-BLS) [5].

Datasets: To evaluate the performance of the proposed ECA-BLS model, extensive experiments are conducted on 30 publicly available binary-class datasets obtained from the UCI Machine Learning Repository [6] and the KEEL repository [7]. These datasets originate from diverse application domains and exhibit substantial variations in sample size, feature dimensionality, and data complexity. Such diversity enables a systematic and comprehensive assessment of the robustness and generalization capability of ECA-BLS across different data distributions and problem scales.

Performance Metrics To rigorously evaluate the effectiveness of the proposed PW-RVFL-CIL framework, its performance is assessed using several widely adopted classification metrics, namely Accuracy, Sensitivity, Specificity, Precision, F-measure, and G-mean. These metrics collectively provide a comprehensive and balanced evaluation of predictive performance, particularly in the presence of class imbalance and



(a) brwisconsin (b) heart-stat (c) monk1 (d) ripley

Fig. 3: Performance variation of the ECA-BLS model with respect to parameters a and b simultaneously.

label noise.

The mathematical definitions of the employed performance measures are given as follows:

$$\text{Accuracy} = \frac{\text{True}_+ + \text{True}_-}{\text{True}_+ + \text{False}_+ + \text{True}_- + \text{False}_-}, \tag{24}$$

$$\text{Sensitivity} = \frac{\text{True}_+}{\text{True}_+ + \text{False}_-}, \tag{25}$$

$$\text{Specificity} = \frac{\text{True}_-}{\text{True}_- + \text{False}_+}, \tag{26}$$

$$\text{Precision} = \frac{\text{True}_+}{\text{True}_+ + \text{False}_+}, \tag{27}$$

$$\text{F-measure} = \frac{2 \times \text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}, \tag{28}$$

$$\text{G-mean} = \sqrt{\text{Sensitivity} \times \text{Specificity}}. \tag{29}$$

These metrics characterize complementary aspects of classification performance. Here, True_+ and True_- denote the number of correctly classified positive and negative samples, respectively, while False_+ and False_- represent false positive and false negative outcomes. The inclusion of F-measure and G-mean ensures a fair and informative evaluation by jointly considering both class-wise performance and balance between sensitivity and specificity, which is particularly important for imbalanced learning scenarios.

S.VII. Win–Tie–Loss (W–T–L) sign test

To further substantiate the statistical significance of the comparative results, we employ the Win–Tie–Loss (W–T–L) sign test [8]. Table I summarizes the pairwise comparisons between the proposed ECA-BLS and the baseline models on the UCI and KEEL datasets. Each entry in the form $[x, y, z]$ indicates that the model listed in the row achieves x wins, y ties, and z losses against the model listed in the corresponding column.

Under the null hypothesis, two competing models are assumed to perform equivalently, implying that each model should win approximately $v/2$ out of v datasets. A statistically significant difference is established if a model achieves at least $v/2 + 1.96\sqrt{v/2}$ wins. In the presence of ties, they are evenly divided between the two models; if the number of ties is odd, one tie is discarded prior to division. For $v = 26$ datasets, the resulting significance threshold is 18 wins.

Based on this criterion, the proposed ECA-BLS achieves at least 18 wins against several strong baselines, including F-BLS, GEIB, and H-ELM, thereby demonstrating statistically significant superiority. Although the threshold is not reached against a few baselines, the overall W–T–L outcomes in Table I consistently favor ECA-BLS. Collectively, these results provide strong evidence that the proposed ECA-BLS model delivers superior and more reliable performance compared to the baseline methods across diverse benchmark datasets.

I. S.VIII. Sensitivity Analysis

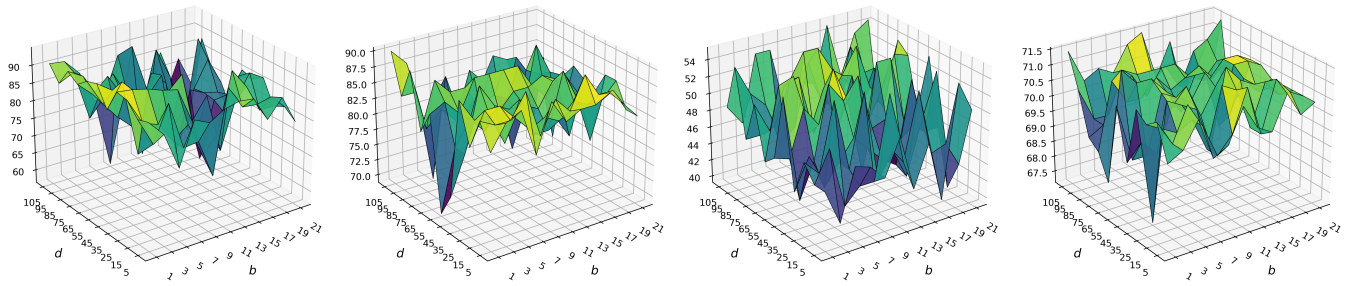
In this section, we investigate the sensitivity of the proposed ECA-BLS model to its key hyperparameters, namely the regularization parameter λ_r and the structural parameters a, b, and d, using four representative datasets: *brwisconsin*, *heart-stat*, *monk1*, and *ripley*.

1) Effect of the regularization parameter λ_r: Figure 2a illustrates the influence of λ_r on classification accuracy. The results indicate that the performance stabilizes when $\lambda_r \geq 0.01$ for all datasets, while excessively large values ($\lambda_r > 1000$) lead to a degradation in accuracy. Accordingly, λ_r is selected from the range $[0.01, 1000]$ in subsequent experiments.

2) Effect of the number of feature groups a: Figure 2b shows the effect of the parameter a on model performance. It can be observed that accuracy improves as a increases and reaches its optimum for $a \geq 15$, after which performance gradually declines beyond $a = 45$. Thus, an appropriate range for a is $[15, 45]$.

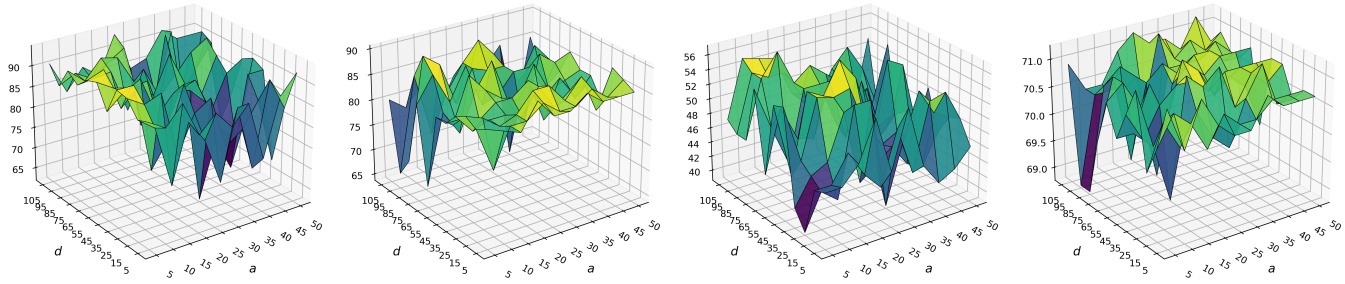
3) Effect of the number of nodes per feature group b: The impact of parameter b is presented in Fig. 2c. The results reveal that optimal performance is consistently achieved when b lies in the range $[7, 11]$, which is therefore adopted in this study.

4) Effect of the number of enhancement nodes d: Figure 2d depicts the sensitivity of ECA-BLS to the parameter d. Unlike the previous parameters, the optimal value of d varies across



(a) brwisconsin (b) heart-stat (c) monk1 (d) ripley

Fig. 4: Performance variation of the ECA-BLS model with respect to parameters b and d simultaneously.



(a) brwisconsin (b) heart-stat (c) monk1 (d) ripley

Fig. 5: Performance variation of the ECA-BLS model with respect to parameters a and d simultaneously.

datasets, indicating that this parameter is inherently dataset dependent.

5) Joint effect of a and b: Figure 3 illustrates the combined influence of a and b on model performance. The results suggest that medium-to-high values of both parameters lead to improved accuracy. Consequently, a and b are selected from the ranges $[13, 21]$ and $[25, 50]$, respectively.

6) Joint effect of b and d: The joint sensitivity of parameters b and d is shown in Fig. 4. The optimal configuration varies across datasets, further confirming the dataset-dependent nature of these parameters.

7) Joint effect of a and d: Figure 5 presents the combined effect of a and d on ECA-BLS performance. The observed optimal regions differ substantially among datasets, indicating that the interaction between these parameters is also dataset dependent.

Since performance varies across datasets and domains, systematic hyperparameter tuning is necessary to achieve reliable generalization.

II. S.IX. Evaluation of the Proposed ECA-BLS under Contaminated Gaussian Noise

To rigorously examine the robustness of the proposed ECA-BLS framework, we conducted experiments on five representative datasets by injecting contaminated Gaussian noise into the training samples. This evaluation aims to simulate realistic noisy learning environments where data corruption and uncertainty are unavoidable.

As reported in Table III, the proposed ECA-BLS consistently demonstrates superior performance compared to the conventional BLS across all considered noise levels and datasets. Notably, ECA-BLS achieves an overall average accuracy of 76.9522%, significantly outperforming BLS, which attains 73.7706%. This clear margin highlights the enhanced noise tolerance of the proposed model.

Furthermore, for each individual dataset and at every noise ratio (ranging from 5% to 40%), ECA-BLS either maintains stable performance or degrades gracefully, whereas BLS exhibits noticeable accuracy deterioration as noise intensity increases. This consistent advantage verifies that the adaptive error-aware correction mechanism embedded in ECA-BLS effectively mitigates the adverse effects of contaminated Gaussian noise.

Overall, these results strongly demonstrate that the proposed ECA-BLS model possesses superior robustness, stability, and generalization capability under noisy conditions. Such characteristics make ECA-BLS particularly well-suited for real-world applications where training data are often imperfect, noisy, or corrupted, thereby establishing its practical superiority over traditional BLS.

TABLE II: Best hyperparameters and performance of the proposed model for the experiments on 26 UCI [6] and KEEL [7] datasets.

Dataset	**(Sensitivity, Specificity, Precision, Recall, F_measure, Gmean, lamda_r, a, b, d)**
acute_inflammation	(100, 100, 100, 100, 100, 100, 0.1, 30, 21, 45)
bank	(0, 100, 0, 0, 0, 0, 100, 10, 1, 25)
brwisconsin	(99.21259843, 69.23076923, 84, 99.21259843, 90.97472924, 82.87680319, 10, 5, 1, 5)
chess_krvkp	(90.6374502, 97.37417943, 97.43040685, 90.6374502, 93.91124871, 93.94544874, 0.1, 20, 19, 75)
cleve	(80.48780488, 89.79591837, 86.84210526, 80.48780488, 83.5443038, 85.01456555, 1, 10, 19, 5)
ecoli-0-1_vs_5	(0, 100, 0, 0, 0, 0, 100, 5, 1, 65)
ecoli-0-1-4-7_vs_2-3-5-6	(15.38461538, 98.86363636, 66.66666667, 15.38461538, 25, 38.99973104, 10000, 5, 1, 85)
ecoli-0-1-4-7_vs_5-6	(28.57142857, 94.62365591, 28.57142857, 28.57142857, 28.57142857, 51.99550967, 100, 5, 1, 75)
ecoli-0-6-7_vs_5	(0, 100, 0, 0, 0, 0, 1000, 5, 1, 95)
haber	(43.75, 86.84210526, 41.17647059, 43.75, 42.42424242, 61.63880357, 0.0001, 25, 1, 45)
heart_hungarian	(77.41935484, 82.75862069, 70.58823529, 77.41935484, 73.84615385, 80.04448152, 1, 25, 9, 45)
heart-stat	(84.7826087, 80, 84.7826087, 84.7826087, 84.7826087, 82.35659473, 1, 30, 19, 15)
ionosphere	(83.75, 80.76923077, 93.05555556, 83.75, 88.15789474, 82.24611284, 0.1, 15, 1, 85)
led7digit-0-2-4-5-6-7-8-9_vs_1	(88.88888889, 91.12903226, 42.10526316, 88.88888889, 57.14285714, 90.00199122, 1, 30, 1, 65)
mammographic	(78.67647059, 86.92810458, 84.2519685, 78.67647059, 81.36882129, 82.69943448, 1, 20, 7, 15)
monk1	(6.493506494, 95.55555556, 55.55555556, 6.493506494, 11.62790698, 24.90964914, 100000, 30, 11, 45)
monk3	(53.65853659, 29.41176471, 42.30769231, 53.65853659, 47.31182796, 39.72646791, 10000, 35, 11, 35)
oocytes_merluccius_nucleus_4d	(88.94472362, 45.37037037, 75, 88.94472362, 81.37931034, 63.52523163, 0.1, 25, 3, 85)
ozone	(0, 100, 0, 0, 0, 0, 10000, 20, 1, 95)
ripley	(63.68421053, 76.75675676, 73.7804878, 63.68421053, 68.36158192, 69.91561669, 0.00001, 5, 5, 25)
shuttle-6_vs_2-3	(66.66666667, 98.48484848, 66.66666667, 66.66666667, 66.66666667, 81.02873913, 10, 50, 15, 65)
spambase	(66.41074856, 98.02325581, 95.31680441, 66.41074856, 78.28054299, 80.68331795, 1, 35, 21, 55)
spectf	(92.30769231, 43.75, 86.95652174, 92.30769231, 89.55223881, 63.54889093, 100, 5, 3, 105)
vertebral_column_2clases	(10.71428571, 95.38461538, 50, 10.71428571, 17.64705882, 31.96839098, 10, 45, 7, 55)
wpbc	(7.692307692, 97.82608696, 50, 7.692307692, 13.33333333, 27.43188585, 10, 5, 1, 25)
yeast-2_vs_4	(0, 100, 0, 0, 0, 0, 1, 5, 1, 25)

TABLE III

Dataset	Noise	ECA-BLS	BLS [1]
ecoli-0-6-7_vs_5	5%	90.9091	90.9091
	10%	90.9091	90.9091
	15%	90.9091	90.9091
	20%	90.9091	90.9091
	30%	90.9091	89.3939
	40%	90.9091	89.3939
Average accuracy		**90.9091**	90.404
ecoli-0-1-4-7_vs_5-6	5%	94	96
	10%	93	92
	15%	93	92
	20%	93	85
	30%	93	90
	40%	93	90
Average accuracy		**93.1667**	90.8333
ecoli-0-1_vs_5	5%	88.8889	87.5
	10%	87.5	86.1111
	15%	88.8889	86.1111
	20%	88.8889	88.8889
	30%	88.8889	76.3889
	40%	88.8889	95.8333
		88.6574	86.8056
monk1	5%	53.2934	46.1078
	10%	47.3054	48.503
	15%	46.1078	52.6946
	20%	47.9042	44.9102
	30%	50.2994	46.1078
	40%	56.2874	46.1078
Average accuracy		**50.1996**	47.4052
vertebral_column_2clases	5%	48.3871	63.4409
	10%	76.3441	60.2151
	15%	34.4086	35.4839
	20%	69.8925	60.2151
	30%	72.043	50.5376
	40%	69.8925	50.5376
Average accuracy		**61.828**	53.405
Overall average accuracy		**76.9522**	73.7706

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